

Marginal populations show physiological adaptations and resilience to future climatic changes across a North Atlantic distribution

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ABSTRACT

Rising levels of anthropogenic carbon dioxide (CO₂) in the atmosphere over the past several decades has resulted in a changing climate and is projected to further fuel global climate change in future centuries. Key components of climate change in the ocean are ocean acidification (decreasing pH and carbonate ion concentration [CO₃²⁻]) and rising sea surface temperatures. While several studies have investigated the effect of these climatic changes on a single population, very few studies have addressed effects on populations living at the margins of their species distribution and the full distributional range. This gap in knowledge impedes the determination of detailed predictions for most species' futures. Over the course of four months, we investigated physiological changes (primary production, respiration, calcification and growth rates) of 6 populations of the intertidal ecosystem engineer and articulated coralline alga *Corallina officinalis* to future climatic conditions (low pH (~7.8); T+3 °C; as well as the combination of low pH and T+3 °C). The populations (n = 2 per geographical location) represent the northern (Iceland) and southern (Spain) margins, as well as the centre (England) of the species distribution in the NE Atlantic. Here, we show that southern and central populations are already living closer to their thermal and stress limits, while Northern populations appear to be the most resilient to environmental changes. We present data confirming light calcification to be the most valuable physiological process which is prioritized in populations throughout the geographical gradient in the NE Atlantic. We found elevated temperature to have a greater effect on populations than pCO₂. Investigating and monitoring organism physiology and structure under these extreme environmental conditions provides important information to predict their acclimatisation and resilience to future environmental conditions and potential changes in their distribution.

1. Introduction

At the forefront of those marine organisms affected by climate change induced impacts, organisms such as calcifying benthic macroalgae act as indicator species (Kroeker et al., 2013b). Among those, the coralline red algae (Corallinales, Rhodophyta) are critical components of polar and tropical marine shallow water ecosystems (Adey and MacInyre, 1973; Steneck, 1986). As ecosystem engineers, they play a key role as a vital structural element of rocky coastal zones (Johansen, 1981; Jones et al., 1994; Nelson, 2009; van der Heijden and Kamenos, 2015; Dayton, 1972; Kelaher et al., 2001; Benedetti-Cecchi, 2006; Noël et al., 2009). They frequently form highly branched, extremely dense and complex turfs which are considered to be at the high end of algal

structural complexity (Coull and Wells, 1983; Davenport et al., 1999) and thus contribute to a wealth of ecosystem services (Hofmann et al., 2020).

Articulated coralline algae (geniculate or upright coralline algae), like *Corallina officinalis*, form dense, carpet-like beds, also referred to as turfs, on hard substrata across large areas in the intertidal ecosystems of the NE Atlantic. *C. officinalis* is generally found in low intertidal, sheltered zones primarily inhabiting the lower part of channels and rock pools that remain damp or filled during extreme conditions or tides, and at the boundary between intertidal to subtidal zones (Digby, 1977; Egilisdottir et al., 2013; Williamson et al., 2014). Articulated coralline algae are thought to have a good ability to adapt to rapid and great changes in environmental conditions, such as physical stress (e.g. wave

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action), water temperature, solar irradiance or carbonate chemistry (e.g. large pH variations) which fluctuate tidally, diurnal, monthly and seasonally (Martone et al., 2010; Egilisdottir et al., 2013; Hofmann et al., 2014; Williamson et al., 2014; McCoy et al., 2016).

Coralline turfs are highly variable, with frond lengths (from the crustose holdfast to the tip of the frond) and density differing at small spatial scales, and host diverse and rich macrofaunal assemblages, capable of supporting 250,000 individuals per m² (Kelaher et al., 2001). In addition, physiological processes such as respiration and photosynthesis alter CO₂ and HCO₃⁻ in the intertidal seawater environment, causing changes in pH across spatial as well as diurnal scales. Changes on both scales are related to species distributions (Morris and Taylor, 1983; Williamson et al., 2014, 2017). The extent of these changes in the carbonate chemistry, particularly in combination with the climatic changes described above, can severely affect coralline algal calcification.

In addition to biologically-induced changes in local carbonate chemistry, intertidal species are exposed to large and frequent fluctuations of seawater levels due to tidal influences. Therefore, as part of the daily cycle, they are exposed to further abiotic and biotic stressors, including direct sunlight, elevated temperatures, exposure to air, fluctuating salinity, strongly fluctuating oxygen and pH levels, wind and wave forces and different predators, including birds (Hampe and Petit, 2005).

Temperature is one of the many key factors influencing the vertical (small-scale) distribution of macroalgae on a shore (Lüning, 1990) and the geographical (large-scale) distribution of macroalgal species (Ganning, 1971). At the organismal level, temperature controls major chemical responses, which consequently affect metabolic pathways (Lobban and Harrison, 1994). For instance, carbonic anhydrase is affected by temperature, varying photosynthetic carbon fixation pathways (Lobban and Harrison, 1994). Water temperature also affects the reproduction, growth, survival and recruitment of macroalgae and accordingly drives species' distributions (Breeman, 1988; Lüning, 1990). In particular, the present increase in global temperatures increases the likelihood of local conditions exceeding species' temperature thresholds, which causes species-level responses such as macroalgal range shifts (reviewed by Helmuth et al., 2006 and Parmesan, 2006; Wernberg et al., 2011; Nicastro et al., 2013; Smale and Wernberg, 2013). Current and future temperatures can cause either acute stress, due to extreme temperature events, or chronic stress, due to gradual warming (Brodie et al., 2014). Adaptations will need to include modifications to all metabolic processes at elevated temperatures, especially those for respiration, photosynthesis, calcification and thus growth, and species may need to respond to both acute and gradual stressors. As ocean conditions continue to change, interactions between variable environmental parameters and coralline algal physiology are expected to be severely affected (Colthart and Johansen, 1973; Hofmann et al., 2012a, b; Egilisdottir et al., 2013; Noiset et al., 2013a; Rendina et al., 2019).

The second key factor causing current changes is ocean acidification (OA), affecting calcareous organisms across their distribution range due to the decrease of calcium carbonate saturation states in the seawater (Orr et al., 2005; Doney et al., 2009). Research has shown coralline algae to be amongst the first calcareous organisms to vanish locally due to acidified seawater (Kleypas et al., 2006; Hall-Spencer et al., 2008). Further studies on crustose coralline algae have shown a combined effect of temperature and elevated pCO₂ that enhances dissolution and affects calcification as well as survival. It is important for these two key factors to be considered in tandem when predicting possible future climate impacts (Martin and Gattuso, 2009; Williamson et al., 2014). Due to all these fluctuations and challenges, coralline algae are thought to be especially vulnerable to OA due to the susceptibility of their high-Mg calcite skeletons, thus reducing survival rates, growth and recruitment (Kroeker et al., 2013b). There is also evidence that coralline algae are able to shift their energy effort from growth extension to

sustaining structural integrity, which is initially weakened under acidified conditions (Ragazzola et al., 2013). Plasticity and variability in skeletal responses (Ragazzola et al., 2013; McCoy and Ragazzola, 2014) and biogeochemical links between photosynthetic and calcification processes may lead to unexpected responses in these organisms, such as a decoupling of physiological processes or a decrease of skeletal inorganic carbon (Borowitzka, 1982; Gattuso et al., 1999; Hofmann et al., 2012a; Johnson and Carpenter, 2012; Kolzenburg et al., 2019).

It remains unclear whether and how the turf-forming, widely distributed coralline alga *C. officinalis* will be capable of withstanding future changes. Based on the overall variability of coralline algal responses discussed above, it is probable that this species will show resilience when exposed to predicted climate change conditions. However, it is uncertain which portions or areas of the species distribution may be most affected. To date, there are very few studies of the physiological responses of benthic intertidal organisms to current and future climate conditions across their full distributional range (reviewed by Helmuth et al., 2006; Díez et al., 2012). This gap in knowledge impedes the determination of detailed predictions for *C. officinalis* and most other species' futures (Nelson, 2009; Brodie et al., 2014; Williamson et al., 2017).

It is proposed that the centre of a species geographic distribution encompasses the most favourable conditions and thus holds the highest population density (Whittaker, 1956; Brussard, 1984). On the margins, compared to the centre of the distribution, conditions are believed to become less favourable owing to increased interspecific competition and greater abiotic stress (Aitken et al., 2008), causing a reduction in population densities and a decline in relative fertility (Case and Taper, 2000). Additionally, lower recruitment than mortality rates have been reported for local marginal – or “sink” – populations, whose survival relies on the influx of spores or zygotes from source populations (Watkinson and Sutherland, 1995).

Here, we investigated physiological responses (respiration, photosynthesis, dark and light calcification, growth rates) of *C. officinalis* to future climatic changes across its natural distribution in the North East Atlantic. Central populations were sampled from England at site with confirmed genetic data by Brodie et al. (2013). Marginal populations were sampled in Spain (southern populations; Williamson et al., 2015) and in Iceland (northern populations; Yesson et al., 2018).

We hypothesised that central populations would be more robust to lower pH levels and elevated temperatures and therefore would be able to acclimatize to the predicted level of acidification and temperature increase. In contrast, we predicted that southern populations, already near thermal limits, would be negatively influenced under higher temperature and that northern populations might be most influenced by lower pH levels due to the increased solubility of CO₂ in colder waters. We expected all populations to be negatively influenced by the treatment combining lower pH and elevated temperature.

2. Material and methods

2.1. Specimen collection

In July 2016, *C. officinalis* populations (n = 2) were collected intertidally (n = 36 per population, total n = 72), during low tide at an average depth of 0.31 m from the water surface at sites in the southeast coast of the UK (CP; central population 1: St Margarets Bay: 51.148056 °N, 1.385056 °E; central population 2: Westbrook Bay: 51.388840 °N, 1.367170 °E; beeline distance between populations: 26.85 km), the northwest coast of Spain (SP; southern margin population 1: Illa del Arousa: 42.56870 °N, 8.89171 °W; southern margin population 2: Tragove: 42.52444 °N, 8.82772 °W; beeline distance between populations: 7.19 km) and the southwest coast of Iceland (NP; northern margin population 1: Stafnes South: 63.968444 °N, 22.750861 °W; northern margin population 2: Stafnes North: 63.974380 °N, 22.753610 °W, beeline distance between populations: 1.59 km). Specimens were

transported into the lab at the Institute of Marine Sciences of the University of Portsmouth, UK using temperature-insulating containers. Only healthy specimens without epiphytes and indication of bleaching or damage were selected for this study. The species was confirmed using Microsatellite analysis as published in [Tavares et al. \(2018\)](#).

2.2. Experimental design

After initial acclimatisation of 4 weeks to laboratory/aquarium conditions, the collected populations ($n = 6$, $n = 2$ per region) were colour coded and exposed to four treatments:

- (1) Control (ambient temperature, ambient $p\text{CO}_2$ (pH 8.1–8.2/380–400 ppm $p\text{CO}_2$, [IPCC, 2014](#)),
- (2) high temperature: T+3 (high temperature (T+3; [IPCC, 2014](#)), ambient $p\text{CO}_2$),
- (3) high $p\text{CO}_2$: CO_2 (ambient temperature, high $p\text{CO}_2$ (pH 7.75–7.80/~1000 ppm $p\text{CO}_2$; RCP 8.5, [IPCC, 2014](#)),
- (4) climate change: T+3/ CO_2 (high temperature, high $p\text{CO}_2$) and cultured for four months.

Fifty 11.1 L aquaria were set up in a flow-through, open seawater system (Figs. S1 and S2). Each was supplied with an individual water supply, a water pump and an adjustable air diffuser, to circulate and mix seawater to obtain an even temperature as well as imitating movement and air saturation most likely experienced in the natural environment. Aquaria were insulated from one another using Styrofoam in order to minimise temperature loss and exchange with the environment and between replicates of different treatments. Wastewater of the replicate aquaria was treated with UV light (Tropic Marine Centre (TMC) P2-110W Commercial UV Steriliser, max. flow rate: 65 L/min) prior to disposal. Each of the four water source tanks (74 L) was also equipped with a water pump for mixing purposes and had twelve individually controllable outlets. The header tanks were temperature controlled by up to 3 chillers (TK-2000 chillers, cooling capacity: 870 W, 800 L/h, TECO) to maintain the desired temperature. Tubes for individual water supply to the replicate aquaria were individually insulated using foam pipe insulation and a radiator heat reflector roll to ensure a minimised temperature loss and exchange with the environment from the water source tanks to the aquarium. Carbon dioxide treatments were supplied with CO_2 enriched pressure air with air tubes, in the identical way to which the ambient $p\text{CO}_2$ treatments were supplied with pressure air. The CO_2 enriched air was constantly produced by mixing pure CO_2 (VK cylinder, BOC, Portsmouth, UK) with pressure air in a mixing flask (5 L) prior to distribution to the treatments. The mixture was consistently monitored with a gas analyser (LI-820, LI-COR Environmental, NB, USA) and a preconnected flowmeter (Rotameter 1.2 L/min, Caché Instrumentation, Wakefield, UK) throughout the experiment to ensure the correct concentration of $p\text{CO}_2$ supply to the experimental aquaria. The fifty aquaria were randomly assigned to a treatment ([Cornwall and Hurd, 2015](#)) and were subdivided into treatments (Figs. S1 and S2). Two additional aquaria solely containing flint stones, which were used to fix the specimens on, were set up alongside the experiment, one with ambient air supply, one with CO_2 enriched air supply. These were added in order to monitor potential influences of the flint stones to the seawater chemistry and were treated like the other aquaria.

Upon arrival in the laboratory, specimens (total rocks with specimens per population and treatment replicate: $n = 4$) were kept at their *in situ* temperatures measured during sampling (northern margin temperature: 11.0 °C; central temperature: 16.0 °C; southern margin temperature: 18.5 °C). The system was set to a light/dark cycle of 9/15 h and an intensity of 230 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ (taken from the average of natural conditions of northern, central and southern locations) produced and controlled by TMC AQUABEAM 600 Ulma Reef White lights with cloud function. Additionally, sunset and sunrise were mimicked through a slow increase and decrease of light intensities over a period of 1 h. All

specimens were kept in their corresponding conditions for a minimum of 1 week to acclimate to aquaria conditions. Following the initial acclimatisation, over the next 3 weeks, one quarter of all, the NP, CP and SP, were gradually acclimatised (increase of 1.0 °C/week and increase or decrease of 50 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ /week, respectively) to elevated temperatures, a further quarter of all populations was gradually acclimatised to elevated $p\text{CO}_2$ (increase of 200 ppm $p\text{CO}_2$ /week which is equivalent to a decrease in pH of ~0.1) and light conditions. A further quarter of all populations was gradually acclimatised to the combination of elevated temperature and $p\text{CO}_2$ as described above and the remaining 25 % of all populations were retained in their initial, Control conditions. After acclimatisation, specimens were randomly and equally distributed within the corresponding replicate aquaria.

2.3. Monitoring of water parameters

Temperature, pH and salinity were measured daily in each replicate, and total alkalinity (A_T) was measured daily in one randomly chosen replicate of each treatment. Irradiance was measured once a month to monitor the decrease in light intensity using a HOBO UA-002-64 Pendant Temp/Light data logger (accuracy: 0.47 °C, resolution: 0.1 °C. Tempcon, Arundel, UK). Temperature and salinity were measured with a CO310-1 portable salinity and temperature probe (accuracy: 0.2 % for salinity, ± 0.2 °C for temperature; resolution: 0.1 for salinity, 0.1 °C for temperature. VWR, Leicestershire, UK). Dissolved oxygen and pH were measured with an HQ30d portable multi-parameter meter and a luminescent dissolved oxygen (LDO101, accuracy: ± 0.1 mg/L. HACH, Manchester, UK) and pH probe (PHC301, accuracy: ± 0.02 pH, HACH, Manchester, UK) calibrated on the National Bureau of Standards (NBS) scale and converted into total scale values using Tris/HCl and 2-aminopyridine/HCl buffer solutions after [Dickson et al. \(2007\)](#). Total alkalinity (A_T) was determined using the alkalinity anomaly technique after [Smith and Key \(1975\)](#); [Chisholm and Gattuso \(1991\)](#) and [Dickson et al. \(2007, SOP 3b\)](#). Titrations were carried out with an automatic titrator TitroLine 7000 (measurement accuracy: 0.002 ± 1 digit, dosing accuracy: 0.15 %, dosing precision: 0.05–0.07 %. Schott SI Analytics, Mainz, Germany) to ensure minimal variation. As titrant 0.1 N hydrochloric acid (HCl) was used and validated against Certified Reference Material provided by Andrew G. Dickson (Batch 154, Scripps Institution of Oceanography).

Due to the high number of samples, each water sample was sterile filtered and subsequently poisoned with 25 μl of a saturated Hg(II)Cl solution (Mercuric chloride 99.5 %, ACROS Organics, Thermo Fisher Scientific, Geel, Belgium) prior to the titrations for A_T determination. Samples were stored in a cool and dry place pending processing.

2.4. Physiological incubation procedure and measurements

Differences in oxygen concentration, pH and A_T were determined in order to calculate photosynthesis, respiration and calcification rates by measuring oxygen, pH and A_T , respectively, before and after the incubations. Prior to A_T measurements, seawater samples were filtered through a syringe filter (hydrophilic 25 mm, 0.2 μm PTFE. Fisherbrand, Loughborough, UK) into sterile 50 ml tubes (Plastic Centrifuge Tubes. Fisherbrand, Loughborough, UK). Immediately after, potentiometric titrations were performed using an automatic titrator calibrated on the NBS scale. Results were calculated based on a Gran function after [Dickson et al. \(2007\)](#). The carbonate system of the seawater was calculated from pH, A_T , salinity and temperature using the Excel Macro CO2Sys (Version 2.1, [Lewis and Wallace, 1998](#)) with K1 and K2 according to [Mehrbach et al. \(1973\)](#) and refit by [Dickson and Millero \(1987\) \(Table 1\)](#). Measurement ranges of pH values represent *in situ* values and are highly dependent on temperature.

To determine changes in photosynthesis, respiration and calcification rates, additional incubations for each population ($n = 6$, replicates each: $n = 3$) of all treatments ($n = 4$) at 230 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ (as

Table 1

Carbonate system parameters during the culturing of *C. officinalis*. All numbers are mean values ($n = 62 \pm \text{SD}$). Salinity, pH, temperature, and total alkalinity (A_T) were measured while other parameters were calculated.

Population	Treatment	pH (NBS scale)	Salinity	Temperature [°C]	A_T [$\mu\text{mol kg}^{-1}$]	DIC [$\mu\text{mol kg}^{-1}$]	HCO_3^- [$\mu\text{mol kg}^{-1}$]	Ω_{Ca}	Ω_{Ar}	pCO_2 [μatm]
Northern	Control	8.19 ± 0.01	35.5 ± 0.29	10.38 ± 0.81	2174 ± 25	1983.9 ± 23.2	1840.0 ± 21.71	3.11 ± 0.08	1.95 ± 0.05	222.2 ± 5.9
	T+3	8.16 ± 0.01	35.6 ± 0.14	13.42 ± 1.86	2156 ± 70	1953.4 ± 67.1	1804.1 ± 63.80	3.26 ± 0.22	2.05 ± 0.14	204.7 ± 18.0
	CO_2	7.85 ± 0.03	35.6 ± 0.03	11.32 ± 1.42	2308 ± 318	2236.8 ± 305.8	2132.1 ± 289.62	1.65 ± 0.37	1.04 ± 0.23	570.4 ± 85.8
	$\text{CO}_2/\text{T}+3$	7.74 ± 0.09	35.7 ± 0.21	13.93 ± 1.42	2357 ± 194	2290.2 ± 193.6	2183.1 ± 186.03	1.64 ± 0.34	1.03 ± 0.21	619.4 ± 156.9
Central	Control	8.18 ± 0.01	35.7 ± 0.24	15.33 ± 1.44	2182 ± 30	1960.3 ± 26.4	1800.4 ± 29.97	3.54 ± 0.13	2.23 ± 0.08	187.4 ± 6.8
	T+3	8.13 ± 0.02	35.8 ± 0.22	18.95 ± 1.39	2197 ± 21	1961.4 ± 14.1	1793.1 ± 15.36	3.75 ± 0.22	2.36 ± 0.14	175.9 ± 12.4
	CO_2	7.82 ± 0.10	35.7 ± 0.30	15.88 ± 1.53	2191 ± 32	2105.6 ± 22.0	2001.2 ± 26.49	1.77 ± 0.42	1.11 ± 0.26	489.9 ± 134.2
	$\text{CO}_2/\text{T}+3$	7.79 ± 0.03	35.8 ± 0.33	19.53 ± 1.38	2198 ± 32	2110.8 ± 26.5	2007.4 ± 24.99	1.77 ± 0.17	1.11 ± 0.11	470.6 ± 48.9
Southern	Control	8.15 ± 0.01	35.8 ± 0.27	18.70 ± 1.19	2194 ± 94	1955.3 ± 78.5	1785.4 ± 65.78	3.80 ± 0.31	2.39 ± 0.20	172.0 ± 2.6
	T+3	8.15 ± 0.01	35.9 ± 0.31	20.85 ± 0.45	2195 ± 73	1944.2 ± 66.0	1767.5 ± 59.05	3.97 ± 0.19	2.50 ± 0.12	161.4 ± 5.2
	CO_2	7.80 ± 0.06	35.8 ± 0.30	18.77 ± 1.21	2152 ± 54	2069.9 ± 36.0	1968.9 ± 31.15	1.69 ± 0.31	1.07 ± 0.19	483.1 ± 95.9
	$\text{CO}_2/\text{T}+3$	7.77 ± 0.05	35.9 ± 0.32	20.92 ± 0.41	2193 ± 84	2121.3 ± 83.2	2020.5 ± 81.65	1.60 ± 0.35	1.01 ± 0.22	541.5 ± 106.4

average of all sites) were performed before and after the study. For this, algal fronds with a fresh weight of 1.04 ± 0.28 g were transferred into a clear closed cell 58 ml incubation chamber and replicates were incubated simultaneously and bubble-free for one hour (AQUARAY Nature Perfect, TMC, London, UK). To account and correct for metabolism effects by other organisms, reference incubations without algae were performed each time alongside the above-described incubations. Aluminium foil was used to coat the chambers to determine calcification and respiration rates in the dark. Irradiance was measured with a Quantitherm PAR/Temperature Sensor with a QTP1 probe (resolution: $1 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, 0.02°C , respectively. Hansatech, Norfolk, UK).

The weight of algal fronds used in the incubations was transformed from FW into dry weight (DW) by multiplying FW measurements with the factor 0.62286. This factor was determined from FW and DW weight measurements of thirty 1 g frond bundles from all populations before and after algae were left to dry for 48 h in an oven at 60°C (Egilsdottir et al., 2013).

Due to the fragility of one of the SP, only one out of two populations was incubated for physiological measurements, but both were maintained and analysed for growth rate analysis.

2.5. Monthly growth rates and death rates

Before the start of the experiment and after the acclimatisation phase, all specimens were stained with a solution of 0.5 g L^{-1} of the vital stain Alizarin Red S (Fluks, Sigma-Aldrich, Steinheim, Germany), following Andrade and Johansen (1980) and Ragazzola et al. (2012). Algae were submerged in the solution for 24 h under a light intensity of 1990 lx ($\sim 35 \mu\text{mol photons m}^{-2} \text{s}^{-1}$) to ensure the incorporation of the stain in the skeleton without exposing the specimens to unnecessary irradiation stress. After staining, algae were carefully and thoroughly rinsed with filtered seawater in order to remove the residual staining solution and placed back into the respective aquaria. After the experiments, specimens of each population and treatment were carefully dried and growth rates were determined by preparing longitudinal sections using sandpaper of two different grit sizes consecutively (Granulation: 180 and 1200, Silicon carbide sanding sheet waterproof T489, NORTON Abrasives, Stafford, UK). Prior to sanding the thalli very carefully with a constant light pressure, thalli were exposed to a 5% diluted solution of household bleach for 3 min (Domestos Pink Bleach, Unilever UK Ltd, Leatherhead, UK) and rinsed with purified water to reduce a potential impact of the bleach on the skeleton (Loxton et al., 2017). Following the bleaching and sanding, thalli were viewed and photographed under a stereo microscope (S8 APO, Leica Microsystems, Milton Keynes, UK) with an attached microscope camera (MC120 HD, Leica Microsystems, Milton Keynes, UK) under $1\times$ magnification. Where stain was visible, growth was measured from the stain to the tip of the main branch using the software ImageJ (Rasband, 2016) with the scale bar set to $1 \text{ mm}/193$

pixels, which was determined using scale paper under the same magnification.

Analysis consisted of replicate specimens for NP, CP and SP, respectively, under Control ($n = 42, 50$ and 65), elevated temperatures ($n = 42, 52$ and 44), pCO_2 ($n = 45, 40$ and 46) and temperature/ pCO_2 ($n = 48, 55$ and 60).

2.6. Data analysis

Data analyses were run in SPSS Statistics 24 and 25 (IBM Corp, 2016 and 2018, respectively) and data were tested for normality and homogeneity prior to analyses (Shapiro-Wilk and Levene Test).

Data for net production, respiration, light calcification and dark calcification ($n = 3$ for all) were normally distributed and showed homogeneity of variance. This resulted in statistical analysis using the parametric two-way analysis of variance (ANOVAs) for individuals from each country of origin (NP, CP and SP ($n = 2$ each)) with treatment (4 levels: Control, T+3, CO_2 , T+3/ CO_2 conditions) as the fixed factor. When significant effects were found, data were further explored by running a post hoc Tukey HSD test. Tests were performed for central and southern populations under each temperature and pCO_2 condition comparing measurements after exposure to treatments and Control.

Data for growth rates were not normally distributed and not homogeneous and were analysed using independent-sample Kruskal-Wallis H tests with pairwise comparison ($n = 39$).

3. Results

The experimental design allowed for very stable parameters under all conditions and did not influence the physiological behaviour of specimens (Table 1). Saturation states and pH for calcite and aragonite were highest in warmer conditions. Flint stones in their respective treatments did not have an effect on the carbonate chemistry of the experiment (Table S3).

3.1. Primary production

No statistically significant differences were detected between populations in each treatment. However, a clear trend towards lower and more variable primary production rates was observed in NP compared to CP and SP.

Within northern populations (NP), NP1 showed higher, though not significantly different, rates in both T+3-included treatments compared to the other two treatments (Fig. 1, Table S1). However, under both CO_2 -included treatments, rates of NP1 declined marginally significant compared to the T+3 treatment (Table S1(B)). NP2 showed little variability in production rates in all but the CO_2 treatment, in which NP2 showed a decline in photosynthetic rates similarly to NP1 (Table S1(A)).

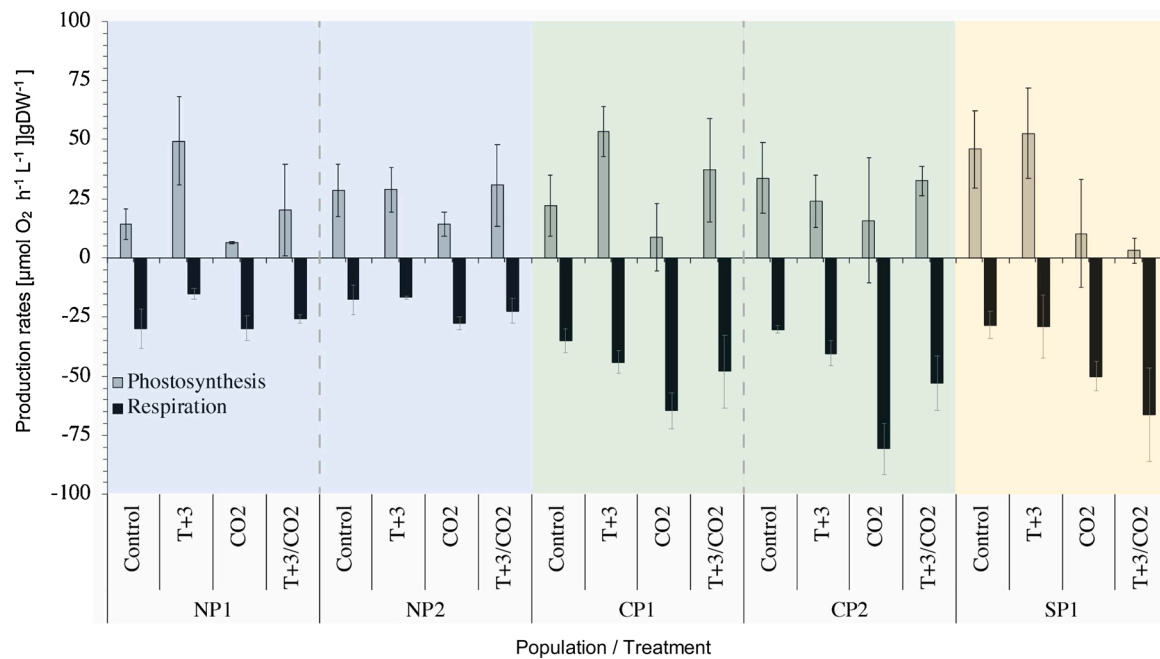


Fig. 1. Gross primary production and respiration rates [$\mu\text{mol h}^{-1} \text{ L}^{-1} \text{ gDW}^{-1}$] $\pm$ SE of *C. officinalis* of each population under each treatment condition. Northern, central and southern populations are abbreviated as NP, CP and SP and represented in blue, green and yellow, respectively. Grey bars represent primary production rates. Black bars represent respiration rates. Dashed lines separate populations of the same region of origin.

No significant difference was found in the Controls or other treatments between both NP.

Central population (CP) 1, like NP, showed higher photosynthetic rates in both elevated temperature treatments compared to their respective Control treatment (Table S1(A)). A significant difference was found in CP1 T+3 vs. CO₂ (Table S1(B)). Both CO₂-elevated treatments showed a decrease in photosynthetic rates for CP1 compared to their respective Control. CP2, however, showed lower but not significantly different rates in T+3 compared to the Control (Fig. 1, Table S1(A)). CP2 showed lower rates in the CO₂ compared to the T+3/CO₂ treatment as well as higher variability (Table S1(A)). No significant difference was found in the Controls or other treatments between both CP.

Due to the lack of material from southern population 2 caused by an already great effect of the treatments on the specimens resulting in fragility and death, only SP1 was incubated for physiological measurements. Control and T+3 treatments in SP1 are higher compared to both CO₂-included treatments (Fig. 1, Table S1(A)). Under treatment T+3/CO₂, SP1 showed the lowest values of all populations in all treatments. In treatment T+3, SP1 showed highest rates within all treatments and of all populations. A significant difference for SP1 was therefore found between Control and T+3 with the T+3/CO₂ treatment (Table S1(B)).

3.2. Respiration

In NP1, rates were highest in the Control and CO₂ treatments (Table S1(A)) and lowest in both temperature-included treatments, with T+3 showing the lowest values (Fig. 1, Table S1(A)). Treatments including elevated temperatures were lower compared to their respective Control treatment. In NP2, less pronounced changes were observed, however, the highest rates were observed in both CO₂-included treatments, with highest rates in the CO₂ treatment and lowest values observed in the T+3 treatment (Fig. 1, Table S1(A)). No significant difference was found in the Controls or other treatments between both NP.

Of CP, both populations showed the same pattern with highest respiration rates found in the CO₂ treatment, followed by the T+3/CO₂ treatment, followed by the T+3 treatment and the lowest rates shown in

Control treatments (Fig. 1, Table S1(A)). Significant differences were found in CP1 CO₂ vs. Control and in CP2 CO₂ vs. Control, T+3, T+3/CO₂ (Table S1(B)). No significant difference was found between the Controls of both CP.

SP1 was the only population where highest respiration rates were observed in the climate change treatment (T+3/CO₂; Fig. 1, Table S1(A)). All other treatments follow the same pattern of CP, with higher rates in the CO₂ treatment, followed by the T+3 treatment, and followed by the Control treatment. Significant differences were found in T+3/CO₂ vs. Control and T+3 (Table S1(B)).

Both NP showed the lowest respiration values in T+3 treatments, whereas all CP and SP showed lowest values in Control treatments, followed by T+3 treatments. All populations showed the highest respiration rates in both CO₂-included treatments, with only SP1 showing the highest rates in T+3/CO₂ and all others in the CO₂ treatments (Table S1(A)). Significant differences were found in: T+3: NP1 vs. both CP; CO₂: both NP vs. both CP, SP1 vs. CP2; T+3/CO₂: both NP vs. CP2 and SP1, NP2 vs. CP1 (Table S1(B)). Respiration rates (Fig. 1) were lowest in NP compared to all other regions of origin, CP and SP showed comparable rates (Table S1(A)). Both NP showed an indistinct yet similar response scheme between all treatments within the location, but a different response scheme compared to CP and SP. SP1 showed the highest and NP showed the lowest respiration rate under climate change (T+3/CO₂) conditions followed by CP. Only in NP did rates of the climate change treatment not differ from rates of the Control.

3.3. Light calcification rates

Calcification rates were highly variable throughout all treatments. Light calcification rates were low in all populations in most treatments, with all populations showing dissolution under different conditions (Table S1(A)). Comparing populations within a treatment, SP1 was significantly different to both NP in all treatments (Table S1(B)) and to both CP in all but the CO₂ treatment (Table S1(B)).

In NP2, calcification rates in the light (Fig. 2, Table S1(A)) were significantly lower under the T+3 treatment compared to all other treatments (Table S1). In both NP, both elevated-CO₂ treatments showed

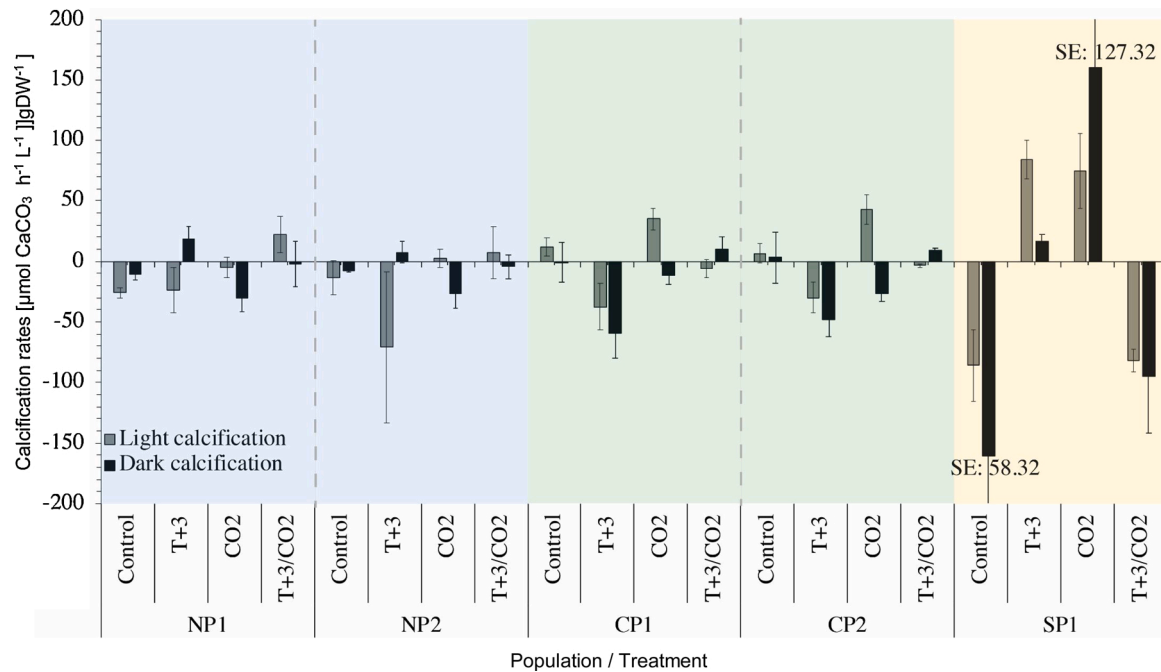


Fig. 2. Light calcification and dark calcification rates [$\mu\text{mol h}^{-1} \text{L}^{-1} \text{gDW}^{-1}$] $\pm$ SE of *C. officinalis* each population under each treatment condition. Northern, central and southern populations are abbreviated as NP, CP and SP and represented in blue, green and yellow, respectively. Grey bars represent light calcification rates. Black bars represent dark calcification rates. Dashed lines separate populations of the same region of origin.

higher light calcification rates than the respective Control treatment ($\text{T+3/CO}_2 > \text{CO}_2 > \text{T+3} > \text{Control}$ and $\text{T+3/CO}_2 > \text{CO}_2 > \text{Control} > \text{T+3}$ for NP1 and NP2, respectively). The Controls showed comparable and not significantly different values between both populations. Comparison between populations showed lower rates of light calcification in NP1 than NP2 in treatments T+3, CO₂, T+3/CO₂ and higher rates in NP1 compared to NP2 in the Control treatment, amounting to a larger overall reduction in light calcification in NP1 under those treatment conditions.

In CP, both populations showed the same response pattern to the different treatments with the highest light calcification rates under CO₂ (Table S1(A)) and the lowest rates under T+3 treatments (Fig. 2; Table S1; $\text{CO}_2 > \text{Control} > \text{T+3/CO}_2 > \text{T+3}$). All elevated-CO₂ treatments showed higher rates than the respective Control treatment (Table S1(A)). The only significant difference was found between the T+3 and CO₂ treatment in CP2 (Table S1(B)). Comparison between populations showed higher rates of light calcification in CP1 than CP2 in treatments Control, T+3, T+3/CO₂ and lower rates in CP1 compared to CP2 in the CO₂ treatment. No significant difference was found in the Controls or other treatments between both CP.

SP1 showed the highest and the most pronounced light calcification rates of all populations in all treatments (Fig. 2, Table S1(A)). In the Control and the climate change treatment, light calcification rates were negative and not significantly different from each other (Table S1). Rates in T+3 and CO₂ treatments were significantly higher than in the Control and climate change treatment and positive; no significant difference was found between the T+3 and CO₂ treatments (Table S1(A)). However, a significant difference between Control vs. T+3 and vs. CO₂ as well as climate change treatment vs. T+3 and vs. CO₂ was found (Table S1(B)).

3.4. Dark calcification rates

Calcification rates in the dark (Fig. 2, Table S1) showed a similar variability to those measured in the light. Both northern populations showed the same response pattern between treatments, with only the CO₂-treatment showing higher rates than the Control (Table S1(A); CO₂

$> \text{Control} > \text{T+3/CO}_2 > \text{T+3}$). Comparison between populations showed higher rates of dark calcification in NP1 than NP2 in treatments Control, T+3, CO₂ and lower rates in NP1 than NP2 in T+3/CO₂. No significant difference was found between the Controls of both NP. A significant difference between Control vs. T+3 and T+3 vs. CO₂ and T+3/CO₂ was only found in NP2.

In CP, as in NP, an identical response pattern between treatments was observed with significantly lowest, negative dark calcification rates in the T+3 and highest, positive rates in the T+3/CO₂ (climate change) treatment (Table S1(A); $\text{T+3/CO}_2 > \text{Control} > \text{CO}_2 > \text{T+3}$, Fig. 2). Comparison between populations showed slightly lower rates of dark calcification in CP1 than CP2 in Control and CO₂ treatments, higher rates in the T+3 treatment, and similar rates in the T+3/CO₂ treatment. No significant difference was found between Controls of both CP. Significant differences were only found in T+3 vs. CO₂ in both CP.

In SP1, lowest, negative rates were found in the Control treatment, followed by higher, negative rates in the T+3/CO₂ (climate change) treatment; higher, positive rates in the T+3 treatment and the highest, positive rates in the CO₂ treatment (Fig. 2, Table S1(A); $\text{CO}_2 > \text{T+3} > \text{T+3/CO}_2 > \text{Control}$). Significant differences were found in SP1 Control vs. T+3 and vs. CO₂, T+3 vs. CO₂ and vs. T+3/CO₂ and CO₂ vs. T+3/CO₂ (Table S1(B)). Additional significant differences between SP1 and other populations were found within treatments: Control: SP1 vs. both NP and vs. both CP; CO₂: SP1 vs. both NP and vs. both CP; T+3/CO₂: SP1 vs. both CP (Table S1(B)).

3.5. Linear growth rates

Monthly growth rates of NP increased in all treatments in relation to the Control (Fig. 3, Table S2(A)). The highest increases in NP1 were observed under elevated temperatures, followed by CO₂ treatments. Comparing NP with each other, NP1 showed lower monthly growth rates than NP2 in the Control and CO₂, comparable rates in the T+3/CO₂ treatment and higher rates in the T+3 treatment (Table S2(A); NP1: $\text{T+3} > \text{T+3/CO}_2 > \text{CO}_2 > \text{Control}$; NP2: $\text{T+3/CO}_2 > \text{CO}_2 > \text{T+3} > \text{Control}$). NP1 showed significant different rates in Control vs. T+3 and vs. T+3/CO₂ (Table S2(B)). No significant difference in the Controls between

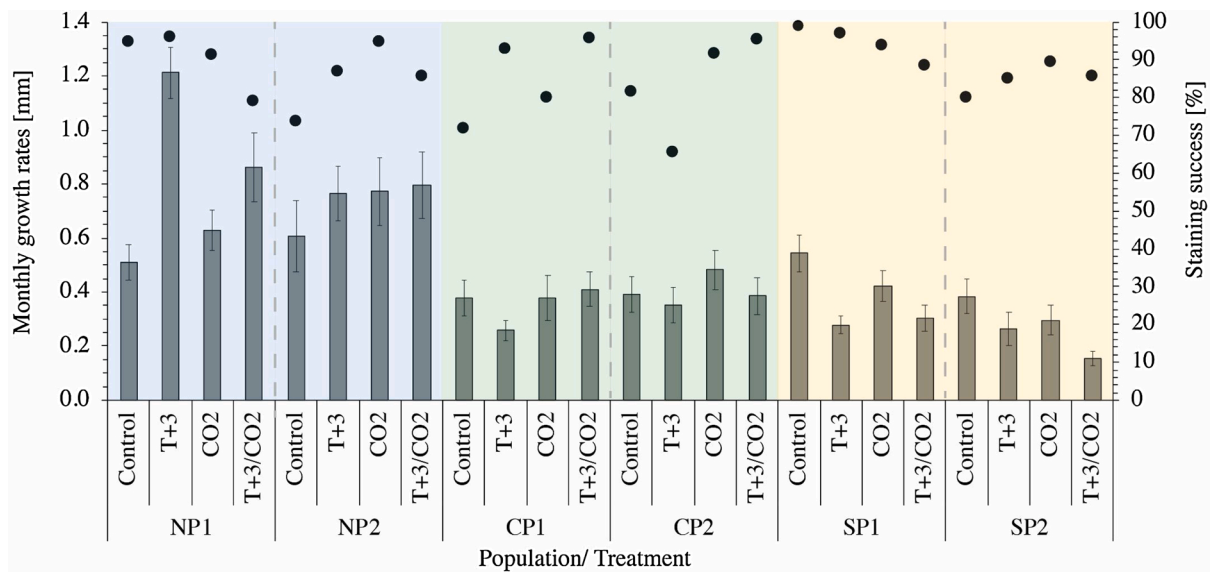


Fig. 3. Monthly linear growth rates [mm month^{-1}] ($\text{AV} \pm \text{SE}$) of *C. officinalis* for northern ($n = 42, 48, 42$ and 45 for Control, T+3, CO_2 and T+3/ CO_2 , respectively), central ($n = 65, 60, 44$ and 46 for Control, T+3, CO_2 and T+3/ CO_2 , respectively) and southern ($n = 50, 55, 52$ and 40 for Control, T+3, CO_2 and T+3/ CO_2 , respectively) populations 1 and 2. Staining success is represented as closed circles ($\text{AV} \pm \text{SE}$, $n = \text{see above}$). Northern, central and southern populations are represented in blue, green and yellow colours, respectively. Dashed lines separate populations of the same region of origin.

both NP was found.

In both CP, growth reduced and showed the lowest rates in the elevated temperature treatment contrary to expectations (Fig. 3, Table S2(A); CP1: T+3/ $\text{CO}_2 > \text{CO}_2 \approx \text{Control} > \text{T+3}$; CP2: $\text{CO}_2 > \text{T+3}/\text{CO}_2 \approx \text{Control} > \text{T+3}$). Comparing CP between each other, CP1 showed lower monthly growth rates than CP2 in the Control, T+3 and CO_2 treatment, but higher rates in the T+3/ CO_2 treatment (Table S2(A)). CP2 showed significant different rates in the CO_2 vs. Control and vs. T+3/ CO_2 comparison (Table S2(B)). No significant difference between the Controls of both CP was found.

In both SP, highest growth rates were found in the Control treatments, followed by high CO_2 treatments (Fig. 3, Table S2(A)). The lowest growth rates were observed in the climate change treatment (Table S2(A); SP1: Control $> \text{CO}_2 > \text{T+3}/\text{CO}_2 > \text{T+3}$; SP2: Control $> \text{CO}_2 > \text{T+3} > \text{T+3}/\text{CO}_2$). Comparing SP between each other, SP1 showed higher monthly growth in all treatments compared to SP2. SP2 showed significant lower rates in the Control vs. CO_2 and vs. T+3/ CO_2 (Table S2(B)).

Comparing populations within treatment, the following significant differences were found: Control: CP2 vs. both NP, SP2; CO_2 : both SP vs. NP1, SP2 vs. CP2, NP1; T+3: both NP vs. both SP and both CP; T+3/ CO_2 : both NP vs. both CP and both SP (Table S2(B), Fig. 3).

Only in NP, growth rates in the treatments T+3 and T+3/ CO_2 were higher than in the Control treatments. Compared between locations, both NP showed the highest growth rates out of all populations and locations as well as the only increase under climate change conditions. Contrary to populations from both marginal locations, CP show no clear pattern of increased or decreased growth rates with treatment. Only in SP, growth rates in CO_2 treatments decreased in comparison to the Control. Marginally lower values but with a more pronounced pattern between treatments of one population were found in SP but not in CP. Most importantly, the only populations decreasing in growth under climate change conditions are both SP.

Staining success always exceeded 65 % in all populations and treatments (Fig. 3). No clear pattern was found associating the staining success with the experimental treatments. The only trend observed was a greater staining success in marginal populations in CO_2 vs. T+3/ CO_2 treatments, compared to the opposite trend in CP.

4. Discussion

This study investigated how physiological traits of *C. officinalis*, from across its geographical distribution in the NE Atlantic, differ when exposed to elevated temperatures and pCO_2 . Due to natural exposure to a wide range of temperatures, *C. officinalis* populations show different, including local and seasonal, adaptation depending on their location. Understanding the impact of future climatic changes on *C. officinalis* and its interactions with the environment is crucial to predict how different lineages of the species and their distributions may be physiologically affected. We present evidence for distinct vulnerability of central and southern peripheral populations of *C. officinalis*, a key marine ecosystem engineer (Kelaher et al., 2003; Daleo et al., 2006), to stress caused by thermal and carbonate chemistry changes. Additionally, we highlight the outstanding resilience of the genetically dissimilar (Yesson et al., 2018) northern populations of *C. officinalis*.

4.1. Within-region assessment

Primary production under climate change conditions increased in all northern populations. Although northern population 1 showed a decrease in respiration rates compared to an increase in northern population 2, elevated temperature and a potential fertilising effect of increased pCO_2 levels in northern populations might contribute to these positive results. An additional reason might be that northern populations already experience acidified conditions (Olafsson et al., 2009) and potential acclimatisation due to the elevated solubility of CO_2 in colder waters (Zeebe and Wolf-Gladrow, 2009). The impact of climatic changes is also observed in slightly reduced light calcification rates, yielding dissolution as well as reduced or stable growth in southern and central populations, respectively. This allows northern populations to emerge as the most resilient populations and region compared to the other tested geographical locations.

Elevated temperatures show a dominating influence on populations over elevated pCO_2 . This result contrasts with a study that found corrosive effects on northern coralline algal populations but only under high CO_2 conditions, not including elevated temperatures (Kamenos et al., 2013). A study on another high-latitude coralline alga, *Clathromorphum nereostrum*, and the encrusting fleshy rhodophyte

Hildebrandia sp. revealed no negative impact on either species by a combination of elevated temperatures and decreased pH (Schoenrock et al., 2016). The results for these two species together with our own results show that these high latitude species, all of them potentially the northern margin of their species distribution, cope exceptionally well with changes in pH and temperature as predicted for the future, despite being farthest from their thermal maxima and optimal pH ranges.

Primary production and respiration rates under climate change conditions increased in all central populations. Linear growth rates of central populations, however, stayed stable, mirroring a simultaneous decrease in light calcification which cannot be compensated for by an increase in dark calcification. Additionally, with values of light calcification surpassing those of dark calcification, light calcification is shown to be the most valuable mechanism and prioritized by the organism. A decrease in deposition of CaCO_3 in the cell walls of the skeleton might be the reason for a sustained and stable growth rate despite the decrease in calcification. This phenomenon has already been described for crustose coralline algae, such as *Lithothamnion glaciale*, *Lithothamnion phymatodeum*, and *Pseudolithophyllum whidbeyense*, by Ragazzola et al. (2013) and McCoy and Ragazzola (2014) where specimens cultured in high CO_2 environments maintained their growth rates but showed a reduction in cell wall thickness, especially those of inter-filament cell walls. Additionally, Kamenos and Law (2010) showed a negative correlation between calcite density and summer temperatures when, compared to lower winter temperatures, in cells of the same species, *L. glaciale*. Thus, it can be concluded that central populations favour spending more energy on linear growth instead of increasing primary production, making these populations less effective in overall productivity.

CO_2 -initiated stress in southern populations was observed under both CO_2 -elevated conditions, expressed by a reduction in photosynthetic rates, an increase in respiration, stable calcification rates and a decrease in growth. These results indicate that southern populations are already at their tolerance limit of acidification stress under current conditions and may not survive the additional, rapid decrease in pH forecasted for the end of this century. Additionally, this result confirms previous evidence for a decoupling between photosynthetic rates and calcification in southern populations that were found under thermal stress conditions (Kolzenburg et al., 2019).

The increase of respiration rates, which were also the highest measured out of all populations, in southern populations combined with low production rates and increased heat stress under climate change conditions indicates that the main driver for stress in these populations is temperature. This has been found for the southernmost populations of another intertidal macroalga, *Fucus serratus* (Pearson et al., 2009). The highly variable response in dark calcification rates in southern populations highlights the importance of including, ideally, more than two populations into future climate change studies to fully understand the response of a region. If great variability truly exists between replicates and therefore individuals, it could also lead to greater resilience to future climatic conditions and could act as a buffer to future environmental conditions at population and possibly species level (McCoy and Widdicombe, 2019). The question arises, however, how the rate of change in environmental parameters suppresses the ability for the optimal adapted individual to reproduce and for this valuable trait to be able to spread amongst the populations under continuously accelerating change.

4.2. Regional comparison

Due to lower photosynthetic rates in both the high CO_2 and climate change treatments compared to the Control in all six populations, it can be concluded that CO_2 may not provide a fertilising effect in *C. officinalis* under a climate change scenario. A similar result was described for tropical calcareous algae, which also highlighted the important difference of an existing CO_2 fertilising effect in fleshy but not in calcareous algae (Johnson et al., 2014).

Compared to central populations and southern populations, northern populations demonstrated lower photosynthetic, respiration, light and dark calcification rates. This is most likely due to temperature effects on metabolism and selection for metabolism to be slower than in central populations or southern populations, which may either have a delayed negative effect on northern marginal populations or, more likely, will provide them with sufficient time to adapt to changing conditions. Additionally, high respiration rates in central populations and southern populations could imply a decrease in stress tolerance due to proximity to the species' stress limits. Although central populations and southern populations showed a reduction in growth and calcification, an associated reduction in primary productivity was only found in southern populations, as previously described in Kolzenburg et al. (2019). This gives rise to the hypothesis that southern populations, already thermally stressed, fail to react appropriately to environmental change, leading to reduced productivity.

Linear growth was also greatest in northern populations compared to central populations and southern populations. Central populations and southern populations showed a reduction of growth in all populations exposed to climate change conditions. This may be explained by a lack of resilience and acclimatisation capacity to stressful future climatic conditions.

Dark calcification in marginal populations is larger in the climate change treatment compared to the Control. Nevertheless, all marginal populations (northern and southern populations) decalcify in the dark, whereas central populations showed positive calcification, leading to the assumption that central populations are more robust towards changes in temperature and carbonate chemistry. Both central populations and southern populations show stable and decreasing growth, respectively, as well as a decrease or relative stability in light calcification rates together with an increase in dark calcification. This shows that even though these populations are trying to compensate the dissolution effect in the light by calcification in the dark, the dissolution effect is too high to be compensated by dark calcification, ultimately weakening not only the capacity for primary production but potentially also the structural integrity of central populations and southern populations.

The experimental design (e.g. average light/dark cycle) allows for the identification of general species reactions to climatic changes, however, conclusions towards how *C. officinalis* would react in their own environment and which environmental influences might affect the physiology of this species have to be drawn with care. In an average light/dark cycle across the species distribution, physiological responses might be overestimated (for southern populations, due to longer light phases in the summer) or underestimated (for northern populations, due to shorter light phases in the summer). However, no photoinhibition was found in this study leading to the conclusion that overestimation is unlikely and despite a potential underestimation, results of all populations already show significant responses under climate change conditions. Marginal populations showed less variability in oxygen production rates, as opposed to CaCO_3 production, under climate change conditions than central populations, which showed greater magnitudes of change. Similar findings arose from a study of a central populations of another coralline algal species, *Ellisolandia elongata*, which showed tolerance to widely fluctuating levels of CO_2 (Egilsdottir et al., 2013). Southern populations showed a different trend than central populations when the climate change treatment was added under field temperature conditions. Thus, under climate change conditions, *in situ* light rather than temperature may influence the response of this species.

Populations are adapted to their respective locations' environmental fluctuations such as temperature and pH. The experimental design is recreating identical treatments for each population to determine the general species responses to those treatments. Therefore, the adaptations to *in situ* natural variability may play an important role when interpreting the results for each population by allowing for greater resilience in populations experiencing naturally higher environmental

fluctuations. With this, the diversity of responses of the species across their latitudinal distribution in the NE Atlantic could be determined.

To conclude, northern populations may be adapting better to warmer temperature conditions and elevated pCO₂ in future oceans. This is very likely due to a smaller temperature range in the field in northern locations during the time of the experiment July - October (10.46 ± 1.82 °C, AV $\pm$ SD) compared to central (17.41 ± 3.79 °C, AV $\pm$ SD) and southern (18.47 ± 2.68 °C, AV $\pm$ SD) locations (Kolzenburg, unpublished data). The broader temperature range and much higher average temperature experienced by specimens from southern populations compared to northern populations could explain why northern populations did not show stress responses in our study, as the thermal exposure remains within their natural thermal range and the temperature initiating thermal stress may not have been reached. Southern populations, however, as described above, already experience the maximum of the species' thermal range and, therefore, show negative physiological responses across the board. This study was planned and conducted under summer conditions, due to thermally reduced metabolic rates in colder temperatures, northern and southern populations might respond with lower rates but possibly similar patterns during winter conditions. Field-based comparisons regarding winter and summer photophysiology of three geniculate coralline algae species (*C. officinalis*, *C. caespitosa* and *E. elongata*), thus of rockpool populations, showed similar patterns for winter and summer measurements in the majority of parameters (Williamson et al., 2014). Southern populations will in all probability surpass their thermal limit under summer conditions within the next decades, leading to major negative physiological responses and probable population extinctions.

Not only may the aforementioned variability within replicates of a population (McCoy and Widdicombe, 2019) have a positive impact on the resilience to thermal stress and this species physiology, but the higher variability of the original location of each population may also be important. Dawes (1989) found a correlation between the thermal variability at the original location site and the performance of the photosynthetic activity, where the fleshy red alga *Eucheuma isiforme* from a very thermal variable environment was able to acclimate much better to temperature changes compared with *E. alvarezii* from a thermal stable environment. However, Noisette et al. (2013b) described somewhat contrasting findings in a study of pCO₂ variability on calcifying red algae of different growth forms and depths. Therein, they described that the articulated coralline alga *E. elongata* living in highly variable and stressful environments, such as the intertidal zone, are less sensitive and more resistant to forecasted pH conditions but may not necessarily benefit from them. Because both studies do not focus on the combined effect of both stressors, and with temperature found to be the major influence in this study, it is likely that *C. officinalis* inhabiting more variable environments, such as the northern margin of its distribution, will be less affected by future thermal and pH changes. Despite giving a good insight in potential resilience, in the context of future climate change our study is considered short-term; thus, future experiments focussing on the exposure of marginal populations to climate change treatments are suggested to last a minimum of six and ideally longer than 12 months. Additionally, light regimes of the respective locations could be introduced to account for not only the length of the light/dark cycle but also for variable light intensities across the species distribution. This will achieve more accurate results as a representation of long term exposure and natural light schemes will allow a more precise interpretation of potential future resilience or adaptation to climate change.

Due to the findings described above, central and southern populations may experience a loss in percentage cover in intertidal, rocky, coastal zones. Central and southern populations' calcification and growth is negatively impacted by elevated sea surface temperatures and ocean acidification, potentially resulting in reduced resilience and subsequently disappearance from within their current habitats and ecosystems. Out of all regions, the physiology of southern populations is

most negatively affected under climate change conditions, suggesting a major deficit of resilience in these populations. With sea surface temperatures warming in northern regions of the species distribution, southern populations may be able to shift their distribution northwards and therefore potentially disappear entirely in their original locations. However, increased dissolution of atmospheric CO₂ in colder waters in the north of the species distribution results in more acidic conditions, forcing species to shift their distributions southwards. This opposing trend diminishes the northernmost and southernmost areas of a species range, forcing species into a narrow band of prevailing viable conditions and resulting in a distribution squeeze in the NE Atlantic.

This study showed the acclimatization potentials of regional populations of *C. officinalis* to predicted future temperatures and pH across its geographic distribution in the NE Atlantic. It clarified significant differences between populations of different areas and presented an unequivocal North-to-South gradient in their physiology and their response to changing temperature. Northern populations proved to be more resilient to and may benefit from climate change conditions compared to central or southern populations. The severe effects on *C. officinalis* caused by changing temperatures and carbonate chemistry could also lead to changes within their respective ecosystems, due to the important functions these algae would no longer perform resulting in a dramatic drop in biodiversity associated with *C. officinalis*.

Author's contributions

RK and FR planned the experiment with input from SM. RK conducted the experiment and analysis with substantial help from FD. RK led the writing of the manuscript with critical input from SM, FD and FR. The corresponding author RK is taking responsibility for the integrity of the work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.envexpbot.2021.104522>.

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